

M.S. Data Science

2 Years(4 Semesters)

1st Semester

Theory

- Mathematical Foundation for Data Science
- Statistics for Data Science
- Python for Data Science
- Quantum Data Science
- Fundamentals of Database Systems: SQL & MongoDB
- Data Visualization & Story Telling
- Open Online Transdisciplinary Course - Soft Skills
- Indian Knowledge Systems 1

Practical

- Hands-on: Python for Data Science
- Hands-on: Data Visualization & Story Telling

2nd Semester

Theory

- Big Data Processing
- Fundamentals of Machine Learning
- Quantum Machine learning
- Elective – I
 - MLOps
 - Developing Web Applications
- Elective – II
 - Data Engineering with Python
 - Data Engineering using Kafka and Spark Structured Streaming
- Completion of MongoDB from MongoDB University
- Indian Knowledge Systems 2

Practical

- Hands-on: Elective -I
- Hands-on: Elective -II

3rd Semester

Theory

- Artificial Intelligence and Deep Learning
- Cloud Computing - AWS / Azure / GCP
- Elective – III
 - Data Science with PySpark
 - AI and Deep Learning through TensorFlow
- Elective – IV
 - Data Engineering using AWS Data Analytics
 - Data Engineering with Google Big Query & Google Cloud
- Open Elective(MOOCs)
 - Artificial Intelligence in Drug Discovery and Development
 - Responsible & Safe AI Systems

Practical

- Hands-on: Elective –III
- Hands-on: Elective –IV

4th Semester

Capstone Project